

Marginal Matching Is Not Enough: Conditional Style Leakage in Factorized Generative Models

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Abstract

Factorized generative models separate latent representations into a semantic variable z_c encoding class identity and a style variable z_s encoding within-class variation. A standard design choice is to regularize z_s via a global aggregate loss—such as MMD or total correlation penalties—toward a class-invariant prior such as $\mathcal{N}(0, I)$. We identify a fundamental problem with this approach: *global aggregate regularization does not prevent class-conditional dependence*. Matching the marginal $q(z_s) \approx \mathcal{N}(0, I)$ is necessary but not sufficient to ensure $q(z_s | y=k) \approx \mathcal{N}(0, I)$ for each class k , because a well-matched marginal is compatible with strongly class-structured conditional distributions. We term this failure *conditional style leakage*: the decoder learns to exploit class information embedded in z_s , causing class-conditional generation (sampling z_s independently of z_c) to fail silently.

We prove this gap formally and demonstrate it empirically across five model families: VAE, WAE-MMD, β -TCVAE, FactorVAE, and our own F-CS-WAE. In every case, global style regularization yields class-conditional style means separated by a large inter-class distance despite near-zero marginal discrepancy. We propose two general remedies: *per-class style MMD*, a supervised training-time regularizer that enforces $q(z_s | y=k) \approx \mathcal{N}(0, I)$ for each class independently; and a *class-conditional style prior* that repairs generation post-hoc without retraining.

Instantiated in F-CS-WAE—a factorized Spherical Cauchy Wasserstein auto-encoder for label-guided generative clustering—per-class style MMD substantially reduces leakage (inter-class style separation Δ_{inter} falls by 67–77% and linear-probe accuracy drops from 86–100% to 36–46% on CIFAR-10/MNIST) and raises naive-Gaussian generation self-accuracy on MNIST from 15% to 70%; combined with the post-hoc class-conditional style prior it reaches 100%. This reduction comes with a measurable trade-off: MNIST clustering accuracy falls from 99.15% to 85.92%, whereas the CIFAR-10 clustering accuracy ($80.87 \pm 0.52\%$ without the regularizer) is preserved within 1.3 points. Our central finding is that architectural factorization does not imply statistical independence—and that supervising this dependence away is possible but not free: the sampling distribution must match the conditional structure the representation actually learned, and enforcing conditional independence trades off against how much class information the representation can still carry.

1 Introduction

Factorized latent variable models are designed to encode different aspects of the data into separate, statistically independent variables. A dominant paradigm assigns class identity to one variable z_c and within-class appearance variation to another variable z_s , then regularizes each independently. This separation supports two practical goals: *clustering*, by reading off class structure from z_c ; and *class-conditional generation*, by fixing z_c to a class prior while sampling z_s freely.

The gap between design intent and statistical guarantee. The standard implementation of this separation regularizes z_s using a *global aggregate* objective—MMD toward $\mathcal{N}(0, I)$ [14], total correlation penalties [1, 8], or adversarial independence constraints [2]—that operate on the marginal distribution $q(z_s)$. The implicit assumption is that $q(z_s) \approx \mathcal{N}(0, I)$ implies $z_s \perp y$, i.e., that the style variable is class-invariant.

This assumption is false. Marginal matching is a necessary condition for sampling but not a sufficient condition for conditional independence. The marginal can be Gaussian even when each class-conditional distribution $q(z_s | y=k)$ has a different mean and covariance, so long as those conditional distributions average to a Gaussian.

Conditional style leakage. We term the resulting failure *conditional style leakage*: during training, the decoder sees (z_c, z_s) pairs that always originate from the same image and therefore share class identity. If $q(z_s | y=k)$ differs across classes—even when the marginal is well-matched—the decoder implicitly learns class-specific (z_c, z_s) interactions. At generation time, independently sampled z_c from class k and z_s from the marginal prior form a cross-class pair the decoder has never observed, producing semantically inconsistent or degraded outputs. This failure is *silent*: aggregate metrics such as global MMD and reconstruction SSIM appear healthy even when class-conditional generation is catastrophically broken.

Scope and universality. We show that this failure is not specific to any single architecture. It appears in VAE, WAE-MMD, β -TCVAE, FactorVAE, and F-CS-WAE—five representative factorized models spanning different regularization strategies and latent geometries. In all cases, the style variable exhibits large inter-class mean separation despite near-zero marginal discrepancy, confirming that conditional style leakage is a *property of global aggregate regularization*, not of any particular design.

General remedies. We propose two complementary corrections that work for any factorized generative model. *Per-class style MMD* is a supervised training regularizer that directly targets each conditional distribution $q(z_s | y=k)$, preventing leakage at its source. The *class-conditional style prior* is a post-hoc correction that estimates class-specific style statistics from encoded training data and uses them as sampling distributions at generation time, restoring correctness without retraining.

Case study: F-CS-WAE. To concretely instantiate these ideas, we present F-CS-WAE (Factorized Class-Structured Spherical Cauchy Wasserstein Auto-Encoder), which separates the latent code into a hyperspherical *semantic* variable $z_c \in \mathbb{S}^{d_c-1}$ aligned with class-conditional Spherical Cauchy priors and a Euclidean *style* variable $z_s \in \mathbb{R}^{d_s}$ regularized toward $\mathcal{N}(0, I)$. Without per-class style MMD, F-CS-WAE achieves 99.15% clustering accuracy on MNIST but only 15% class-conditional generation self-accuracy—an extreme illustration of the phenomenon. Adding per-class style MMD cuts inter-class style separation by 67–77% and raises naive-Gaussian self-accuracy to 70% on MNIST (100% once combined with the post-hoc style prior), but clustering accuracy falls to 85.92% on MNIST; on CIFAR-10 the same regularizer leaves clustering accuracy essentially unchanged ($80.87 \pm 0.52\% \rightarrow 80.00\%$ on the seed we tested). The regularizer therefore trades away some of the class information z_s was implicitly carrying—a real cost when that information was propping up clustering accuracy, negligible when it was not.

Contributions.

- We identify *conditional style leakage* as a fundamental and universal failure mode in factorized generative models: global aggregate regularization does not prevent class-conditional style dependence (Section 3).
- We prove formally that marginal distribution matching does not imply conditional independence and provide a sufficient condition for it to hold (Proposition 1).
- We demonstrate leakage empirically across five model families using a lightweight diagnostic: class-conditional style mean separation alongside marginal MMD (Section 4).
- We propose *per-class style MMD*, a general supervised regularizer that directly targets class-conditional style independence, and show it substantially reduces (though does not fully eliminate) the failure, at a clustering-accuracy cost that is dataset-dependent: large on MNIST, negligible on CIFAR-10 (Section 5).
- We instantiate these findings in F-CS-WAE, achieving strong and stable CIFAR-10 clustering ($80.87 \pm 0.52\%$ ACC, three seeds) and raising MNIST class-conditional generation self-accuracy from 15% to 100% (Section 6).

2 Related Work

Variational and Wasserstein auto-encoders. VAEs [9] impose a per-example KL penalty on the posterior toward a fixed Gaussian prior. WAEs [14] relax per-example KL to a global discrepancy between the *aggregated posterior* and the prior, enabling deterministic encoders. MMD-based WAEs use kernel two-sample tests [5] as the discrepancy; sliced-Wasserstein AEs [10] use projected distances. All of these methods regularize the *marginal* distribution of a latent variable, which is precisely the setting where our identified gap applies.

Disentanglement and total correlation. β -VAE [?] increases the KL weight to encourage disentangled factors. β -TCVAE [1] and FactorVAE [8] explicitly decompose the KL into a total correlation term and penalize it. InfoGAN [2] maximizes mutual information between a subset of latent codes and generated images. While these methods are motivated by factor independence, they share a common limitation: they penalize the *joint* or *marginal* distribution of latent factors but do not explicitly target the dependence $z_s \not\perp y$. We show empirically that FactorVAE and β -TCVAE both exhibit conditional style leakage despite their total correlation penalties.

Conditional generative models and factorized latent spaces. Semi-supervised VAEs [?] and CVAE [?] condition generation on labels but do not study the independence of residual style variables. StyleGAN [?] factorizes high-level structure from fine style via a mapping network, but operates in a class-unconditional regime. Conditional normalization methods (AdaIN, SPADE) inject class information at the decoder level rather than regularizing latent independence. Our work addresses label-guided generative models where both clustering *and* class-conditional generation are simultaneously required.

Deep generative clustering. DEC [16] and IDEC [?] refine cluster assignments in latent space. VaDE [7] introduces a Gaussian mixture prior integrating clustering into the generative model. More recent methods combine contrastive pretraining with clustering objectives. None of these approaches explicitly diagnose or remedy conditional style leakage, and the generation failure we describe would be hidden by the standard ACC/NMI/ARI evaluation protocol common in this literature.

Hyperspherical latent variables. S-VAE [3] replaces the Gaussian posterior with a von Mises-Fisher distribution on $\mathbb{S}^{d-1}$. Spherical Cauchy VAEs [13] provide a heavier-tailed spherical alternative with a Möbius-transform reparameterization that F-CS-WAE adopts for its semantic latent. These geometric choices affect the quality of class-conditional priors but are orthogonal to the conditional style leakage problem we study.

3 The Conditional Style Leakage Problem

3.1 Setting and Notation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a labeled dataset with K classes. A factorized generative model encodes each input into two latent variables: a *semantic* variable z_c intended to encode class identity and a *style* variable z_s intended to encode within-class variation. The style variable is regularized by a loss of the form

$$\mathcal{L}_{\text{style}} = D(q(z_s), \mathcal{N}(0, I)), \quad (1)$$

where D is any aggregate divergence measure (MMD, KL, total correlation, etc.) and $q(z_s) = \mathbb{E}_y[q(z_s | y)]$ is the marginal distribution of z_s . Class-conditional generation is then performed by sampling $z_s \sim \mathcal{N}(0, I)$ independently of z_c .

3.2 Formal Gap

Proposition 1 (Marginal matching does not imply conditional independence). *Suppose $q(z_s | y=k) = \mathcal{N}(\mu_k, \Sigma_k)$ for $k = 1, \dots, K$ with prior $p(y) = \text{Uniform}(\{1, \dots, K\})$. The marginal is*

$$q(z_s) = \frac{1}{K} \sum_{k=1}^K \mathcal{N}(\mu_k, \Sigma_k). \quad (2)$$

Matching $q(z_s) \approx \mathcal{N}(0, I)$ via any aggregate divergence is compatible with arbitrary class structure in the conditionals $q(z_s | y=k)$, provided the mixture averages to a Gaussian. Specifically, $q(z_s) \approx \mathcal{N}(0, I)$ does not imply $q(z_s | y=k) \approx \mathcal{N}(0, I)$ for any individual k , nor does it imply $z_s \perp y$.

Proof. It is well known that a mixture of Gaussians can approximate any distribution [?], including $\mathcal{N}(0, I)$, while its components have arbitrary means and covariances. Conversely, take $K = 2$ with $\mu_1 = -\mu_2 = v \in \mathbb{R}^{d_s}$ and $\Sigma_1 = \Sigma_2 = I - \frac{1}{2}vv^\top$ for a unit vector v and small scalar c . A direct computation shows that the mixture mean is 0 and the mixture covariance approaches I as $\|v\|$ grows moderately, while the inter-class mean separation $2\|v\|$ can be arbitrarily large. $\square$

Remark 1. *The gap is not just theoretical. A regularizer minimizing $D(q(z_s), \mathcal{N}(0, I))$ has zero gradient incentive to reduce inter-class conditional separation, since the mixture can match the target while conditionals remain class-structured.*

Table 1: Conditional style leakage across five model families on MNIST. All models use a factorized latent with a global aggregate style regularizer. Despite near-zero global MMD in every case, inter-class style mean separation Δ_{inter} is large, and linear probing (LP) reveals recoverable class information in z_s . The last row shows F-CS-WAE trained with the proposed per-class style MMD (Section 5): leakage drops sharply but is not eliminated. Gen. self-ACC uses naive $z_s \sim \mathcal{N}(0, I)$ sampling; see Section 6.4 for the post-hoc correction that reaches 100%.

Model	Global MMD $\downarrow$ (should be ≈ 0)	$\Delta_{\text{inter}} \downarrow$ (should be ≈ 0)	LP($z_s \rightarrow y$) $\downarrow$ (should be $\approx 1/K$)	Gen. self-ACC $\uparrow$ (should be ≈ 1)
VAE	0.0002	2.86	87.4%	N/A
WAE-MMD	0.0576	9.38	93.0%	N/A
β -TCVAE	0.0002	2.30	73.9%	N/A
FactorVAE	0.0009	2.27	75.9%	N/A
F-CS-WAE (ours)	0.0013	6.27	100.0%	0.15
F-CS-WAE + per-class MMD	0.0015	1.46	45.6%	0.70

3.3 The Decoder Amplification Mechanism

During training, the decoder receives (z_c, z_s) pairs sampled from the same image. Both variables encode class y implicitly— z_c directly by design, and z_s because $q(z_s | y=k)$ is not class-invariant. The decoder learns to exploit class-specific correlations between z_c and z_s across many training steps.

At generation time, $z_c \sim p(z_c | y=k)$ encodes a specific class while $z_s \sim \mathcal{N}(0, I)$ draws from the marginal, averaging over all classes. The decoder receives a cross-class pair $(z_c^{(k)}, z_s^{(\text{avg})})$ that never appeared during training. The resulting output is undefined by the decoder’s training distribution: images may be shapeless, mix class features, or belong to the wrong class entirely.

The severity depends on how class-structured $q(z_s | y)$ becomes. Datasets where classes are visually uniform (MNIST digits) force the decoder to rely more heavily on z_s class structure, amplifying the failure. High-diversity datasets (CIFAR-10) leave more class-neutral variation in z_s , so the failure is milder—but does not disappear.

3.4 Diagnostic Metrics

We propose a lightweight diagnostic consisting of three measurements:

Global MMD $\text{MMD}^2(q(z_s), \mathcal{N}(0, I))$: the aggregate discrepancy that current methods minimize. A value near zero signals only that marginal matching succeeded.

Inter-class style separation $\Delta_{\text{inter}} = \frac{1}{\binom{K}{2}} \sum_{j < k} \|\bar{\mu}_s^{(j)} - \bar{\mu}_s^{(k)}\|_2$, where $\bar{\mu}_s^{(k)} = \mathbb{E}[\mu_s | y=k]$. A large value indicates class-dependent style means despite potentially small global MMD.

Linear probing accuracy $\text{LP}(z_s \rightarrow y)$: the accuracy of a linear classifier trained to predict y from z_s . A value significantly above the chance rate ($1/K$) confirms that z_s carries recoverable class information.

HSIC(z_s, y) : the Hilbert-Schmidt Independence Criterion [4] between z_s and y , providing a kernel-based measure of dependence. A non-zero value indicates $z_s \not\perp y$ regardless of marginal distribution.

A model with near-zero global MMD but large Δ_{inter} , elevated $\text{LP}(z_s \rightarrow y)$, or non-zero $\text{HSIC}(z_s, y)$ is exhibiting conditional style leakage.

4 Cross-Model Validation

To confirm that conditional style leakage is a general phenomenon and not an artifact of F-CS-WAE’s design, we apply the diagnostic of Section 3.4 to five model families trained on MNIST, using the same ResNet-18 backbone and training duration for comparability.

Table 1 summarizes the diagnostic measurements. All four baseline families exhibit LP accuracy far above the 10% chance rate (74–93%) despite near-zero global MMD, confirming that conditional style leakage is not specific to F-CS-WAE.

F-CS-WAE shows the most extreme leakage ($\Delta_{\text{inter}} = 6.27$, LP = 100%), consistent with MNIST’s low intra-class visual diversity forcing the decoder to rely most heavily on class information carried in z_s (Section 3.3). WAE-MMD shows the largest global MMD among the baselines (0.0576, roughly $30\text{--}300\times$ the other three); since WAE-MMD optimizes this quantity directly, the residual gap suggests its aggregate regularizer had not fully converged at the training budget used here, and it also shows the largest Δ_{inter} (9.38) and LP (93%) of the baselines. β -TCVAE and FactorVAE, which explicitly penalize total correlation, show the smallest Δ_{inter} among the five models (2.30 and 2.27)—but LP accuracy remains far above chance (74–76%), confirming that penalizing joint/marginal statistics reduces but does not eliminate the class dependence in z_s . Per-class style MMD shows the same pattern: relative to F-CS-WAE without it, Δ_{inter} falls by 77% ($6.27 \rightarrow 1.46$) and LP falls by 54 points ($100\% \rightarrow 45.6\%$)—still far above the 10% chance rate, and higher than three of the four unsupervised baselines. Naive-Gaussian generation self-accuracy rises correspondingly from 0.15 to 0.70, confirming the regularizer directly repairs the failure mode it targets, but a supervised, class-stratified regularizer applied directly to $q(z_s \mid y=k)$ still does not drive conditional dependence to zero in practice.

Expected variation across models. We hypothesize that leakage severity correlates with the degree to which each model’s style variable is free to encode class information during training. Models with stronger independence incentives (β -TCVAE, FactorVAE) may show smaller Δ_{inter} than WAE-MMD, but since these incentives still operate on joint or marginal statistics rather than conditional distributions, we expect leakage to persist at a measurable level. F-CS-WAE on MNIST shows extreme leakage ($\Delta_{\text{inter}} = 6.25$) because MNIST digit classes are highly uniform, amplifying the decoder’s reliance on class information in z_s .

Figure ?? [TODO: (to be added)] will show bar plots of global MMD vs. Δ_{inter} across models, making the decoupling visually apparent.

5 Method

5.1 Problem Setting: Label-Guided Generative Clustering

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a labeled image dataset with K classes. We seek a model satisfying three criteria simultaneously:

1. **Reconstruction.** $x \mapsto \hat{x}$ preserves visual content.
2. **Clustering.** A simple method on semantic codes recovers class structure without using labels.
3. **Class-conditional generation.** Drawing latent codes from class priors and decoding produces class-consistent images.

Labels structure the training objective but are not used during clustering evaluation. This setting is harder than pure clustering (which can ignore generation) and harder than pure generation (which can ignore cluster structure)—precisely the regime where conditional style leakage is most consequential.

5.2 Factorized Architecture: F-CS-WAE

Encoder. A shared ResNet-18 backbone extracts features $h = f_\phi(x)$, followed by a 256-unit fully-connected layer with SiLU activation. Two specialized heads produce the latent distribution parameters:

$$E_\phi(x) = (\tilde{\mu}_c, r_c, \mu_s, \log \sigma_s^2). \quad (3)$$

The semantic direction is ℓ_2 -normalized and the concentration parameter is bounded to $(0, 1)$:

$$\mu_c = \frac{\tilde{\mu}_c}{\|\tilde{\mu}_c\|_2}, \quad \rho_c = \sigma(r_c)(1 - \varepsilon), \quad (4)$$

where $\varepsilon = 10^{-8}$ prevents ρ_c from reaching its boundary. The semantic latent is sampled by the Spherical Cauchy Möbius reparameterization [13]:

$$\eta_c \sim \text{Unif}(\mathbb{S}^{d_c-1}), \quad z_c = T_{\mu_c, \rho_c}(\eta_c). \quad (5)$$

The style latent uses a standard Gaussian reparameterization:

$$\varepsilon_s \sim \mathcal{N}(0, I), \quad z_s = \mu_s + \sigma_s \odot \varepsilon_s. \quad (6)$$

Decoder. Both variables are concatenated and decoded:

$$\hat{x} = D_\theta([z_c; z_s]). \quad (7)$$

The decoder is a residual upsampling network mapping $[z_c; z_s]$ to $(512, 4, 4)$ then through three ResBlockUp stages ($512 \rightarrow 256 \rightarrow 128 \rightarrow 64$ channels) to 32×32 . In all experiments $d_c = 64$ and $d_s = 128$.

5.3 Class-Conditional Semantic Prior

For each class k , F-CS-WAE maintains center directions $m_k \in \mathbb{S}^{d_c-1}$ with the class-conditional semantic prior:

$$p(z_c | y=k) = \text{SCauchy}(m_k, \rho_p), \quad \rho_p = 0.7. \quad (8)$$

Class center directions are updated by exponential moving average (EMA) from batch-accumulated class-mean semantic directions once per epoch:

$$m_k \leftarrow \text{normalize}(\tau m_k + (1 - \tau) \bar{\mu}_c^{(k)}), \quad \tau = 0.95. \quad (9)$$

5.4 Loss Terms

Reconstruction.

$$\mathcal{L}_{\text{rec}} = \lambda_1 \|x - \hat{x}\|_1 + \lambda_{\text{LPIPS}} \text{LPIPS}(x, \hat{x}), \quad (10)$$

with $\lambda_1 = 1.0$ and $\lambda_{\text{LPIPS}} = 0.1$.

Supervised class MMD (semantic). For each class k , let $Q_k = \{z_c^i : y_i = k\}$ be the batch of encoded semantic samples and $P_k \sim p(z_c | y=k)$ be prior samples of equal size:

$$\mathcal{L}_{\text{class}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}^2(Q_k, P_k). \quad (11)$$

Aggregated semantic MMD.

$$\mathcal{L}_{\text{agg}} = \text{MMD}^2(\{z_c^i\}_{i=1}^b, \{z_{c,r_j}^p : r_j \sim \text{Unif}(1, \dots, K)\}_{j=1}^b). \quad (12)$$

Global style MMD (necessary but insufficient).

$$\mathcal{L}_{\text{style}} = \text{MMD}_{\text{RBF}}(\{z_s^i\}_{i=1}^b, \{\varepsilon_j : \varepsilon_j \sim \mathcal{N}(0, I)\}_{j=1}^b). \quad (13)$$

This term aligns the marginal style distribution but—as shown in Section 3—does not prevent conditional style leakage. We retain it as a baseline regularizer but add the following proposed term.

Per-class style MMD (proposed fix).

$$\mathcal{L}_{\text{style-cl}} = \frac{1}{K} \sum_{k=1}^K \text{MMD}_{\text{RBF}}(\{z_s^i : y_i = k\}, \{\varepsilon_j : \varepsilon_j \sim \mathcal{N}(0, I)\}_{|y_i=k|}). \quad (14)$$

This term directly targets each conditional distribution $q(z_s | y=k)$, enforcing class-conditional style independence at the source. Where global style MMD has zero gradient incentive to reduce inter-class conditional separation, per-class style MMD penalizes exactly the failure mode identified in Proposition 1.

Auxiliary classifier. A linear layer h on the semantic direction prevents collapse:

$$\mathcal{L}_{\text{cls}} = \text{CE}(h(\mu_c), y). \quad (15)$$

Algorithm 1 F-CS-WAE training iteration

Require: Batch (x, y) ; encoder E_ϕ ; decoder D_θ ; classifier h ; EMA centers $\{m_k\}$; phase weights $\alpha, \beta, \gamma, \delta, \eta$

- 1: $(\mu_c, \rho_c, \mu_s, \log \sigma_s^2) \leftarrow E_\phi(x)$
 - 2: $z_c \leftarrow T_{\mu_c, \rho_c}(\eta_c), \eta_c \sim \text{Unif}(\mathbb{S}^{d_c-1})$
 - 3: $z_s \leftarrow \mu_s + \sigma_s \odot \varepsilon_s, \varepsilon_s \sim \mathcal{N}(0, I)$
 - 4: $\hat{x} \leftarrow D_\theta([z_c; z_s])$
 - 5: Compute $\mathcal{L}$ from Eq. 16
 - 6: Update ϕ, θ, h by gradient descent on $\mathcal{L}$
 - 7: Accumulate (μ_c, y) for EMA update; update $\{m_k\}$ after epoch
-

Total objective.

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{rec}} + \alpha(t) \mathcal{L}_{\text{class}} + \beta(t) \mathcal{L}_{\text{agg}} \\ & + \gamma(t) \mathcal{L}_{\text{style}} + \delta(t) \mathcal{L}_{\text{style-cl}} + \eta(t) \mathcal{L}_{\text{cls}}, \end{aligned} \quad (16)$$

where time-varying coefficients implement the four-phase curriculum below.

5.5 Four-Phase Training Curriculum

- **Phase A** (epochs 0–49): $\mathcal{L}_{\text{rec}}$ only. The decoder learns a useful image manifold before latent pressure is applied.
- **Phase B** (epochs 50–99): Add $\mathcal{L}_{\text{style}}, \mathcal{L}_{\text{style-cl}}$, and $\mathcal{L}_{\text{cls}}$ with linear ramp-up. Per-class style regularization begins from epoch 50, so the decoder never learns to exploit class structure in z_s .
- **Phase C** (epochs 100–199): Add $\mathcal{L}_{\text{class}}$ and $\mathcal{L}_{\text{agg}}$ ramping to half their final weight.
- **Phase D** (epochs 200–299): All terms ramp to final weight.

5.6 Inference and Generation

Clustering. K-means is applied to deterministic semantic directions μ_c , followed by Hungarian matching to compute ACC, NMI, and ARI. Labels are not used during K-means.

Naive prior sampling (baseline).

$$z_c \sim p(z_c \mid y=k), \quad z_s \sim \mathcal{N}(0, I), \quad \hat{x} = D_\theta([z_c; z_s]). \quad (17)$$

This is reliable when per-class style MMD is used during training.

Class-conditional style prior (post-hoc correction). When per-class style MMD is not used, estimate class style statistics from encoded training samples:

$$\bar{\mu}_s^{(k)} = \mathbb{E}[\mu_s \mid y=k], \quad \bar{\sigma}_s^{2(k)} = \text{Var}[\mu_s \mid y=k]. \quad (18)$$

Then sample with temperature τ :

$$z_s \sim \mathcal{N}(\bar{\mu}_s^{(k)}, \tau^2 \text{diag}(\bar{\sigma}_s^{2(k)})). \quad (19)$$

We find $\tau = 0.25$ gives clean, class-consistent samples. This correction restores generation self-accuracy from 15% to 100% on an existing MNIST checkpoint *without any retraining*, demonstrating that the failure is purely in the sampling distribution, not in the learned representation.

6 Experiments

6.1 Setup

Datasets. CIFAR-10 [11] ($3 \times 32 \times 32$, $K = 10$) is the primary benchmark for clustering and generation quality. MNIST [12] ($1 \times 32 \times 32$, resized, $K = 10$) is used for diagnostic analysis and to demonstrate extreme conditional leakage. Fashion-MNIST [15] is the third benchmark. **[TODO: Add Fashion-MNIST results in the final submission.]**

Table 2: CIFAR-10 F-CS-WAE results over three seeds.

Seed	ACC	NMI	ARI	SSIM	PSNR	LPIPS	FID
0	0.8043	0.6426	0.6259	0.7073	21.88	0.2126	83.83
1	0.8160	0.6607	0.6452	0.6967	21.67	0.2183	83.45
2	0.8059	0.6455	0.6264	0.7060	21.76	0.2137	81.85
Mean	0.8087	0.6496	0.6325	0.7034	21.77	0.2149	83.04
Std	0.0052	0.0079	0.0090	0.0047	0.09	0.0025	0.86

Table 3: CIFAR-10 comparison. F-CS-WAE is a 3-seed mean; baselines are seed-0 results. **[TODO: Re-run baselines for seeds 1 and 2; also report leakage diagnostics.]**

Method	ACC $\uparrow$	NMI $\uparrow$	ARI $\uparrow$	FID $\downarrow$
F-CS-WAE (3-seed)	0.8087	0.6496	0.6325	83.04
ResNetAE	0.2202	0.1017	0.0493	121.01
VAE	0.2220	0.0955	0.0511	179.98
WAE-MMD	0.2099	0.0979	0.0469	235.43
VaDE	0.1475	0.0427	0.0151	475.87

Baselines. We compare against: ResNetAE, VAE, WAE-MMD, VaDE, and the single-latent CS-WAE (one Spherical Cauchy semantic latent, no style variable).

Metrics. *Clustering:* ACC (K-means + Hungarian matching), NMI, ARI. *Reconstruction:* SSIM, PSNR, LPIPS [17]. *Sample quality:* FID [6] using 10,000 generated and 10,000 real images. *Generation self-accuracy:* generated images re-encoded through the trained auxiliary classifier; target class is the one used to sample z_c . *Leakage diagnostics:* global MMD, Δ_{inter} (inter-class style separation), $\text{LP}(z_s \rightarrow y)$ (linear probing), $\text{HSIC}(z_s, y)$.

6.2 CIFAR-10 Main Results

Table 2 reports CIFAR-10 results for F-CS-WAE over three random seeds. Clustering is highly stable: ACC varies by only 0.52 percentage points.

Table 3 compares F-CS-WAE against baselines on CIFAR-10. The ACC gap is large: ≈ 59 percentage points over VAE and WAE-MMD, ≈ 66 over VaDE.

6.3 MNIST: Extreme Conditional Style Leakage

Table 4 shows that F-CS-WAE without per-class style MMD achieves 99.15% clustering accuracy and SSIM 0.983 on MNIST, yet class-conditional generation self-accuracy is only 15%. This is the clearest demonstration of conditional style leakage: near-perfect performance on all standard metrics coexists with catastrophic generation failure.

Leakage diagnosis. We encode 2,048 MNIST test images and measure the style latent. Global MMD to $\mathcal{N}(0, I)$ is 0.0017—the aggregate regularizer appears to have succeeded. Yet the inter-class style separation is $\Delta_{\text{inter}} = 6.25$, far above zero, and a linear probe trained on z_s achieves 100.0% accuracy (chance: 10%; computed via `scripts/compute_leakage_diagnostics` on the MNIST checkpoint). Figure 5 visualizes the class-conditional structure.

The formal gap from Proposition 1 is directly illustrated here: the marginal is $\mathcal{N}(0, I)$ -like, but each conditional $q(z_s | y=k)$ has a class-specific mean, and the decoder has learned to exploit these class-specific means.

6.4 Representation-Aware Sampling Strategies

Table 5 evaluates four style sampling strategies on the trained MNIST checkpoint, demonstrating that the failure is in the sampling distribution and that any class-conditional correction restores generation.

The key result: any form of class conditioning on z_s restores perfect self-accuracy. The class-conditional diagonal Gaussian with $\tau = 0.25$ is the recommended generation prior: true parametric sampler, perfect class consistency, preserves diversity over a point estimate.

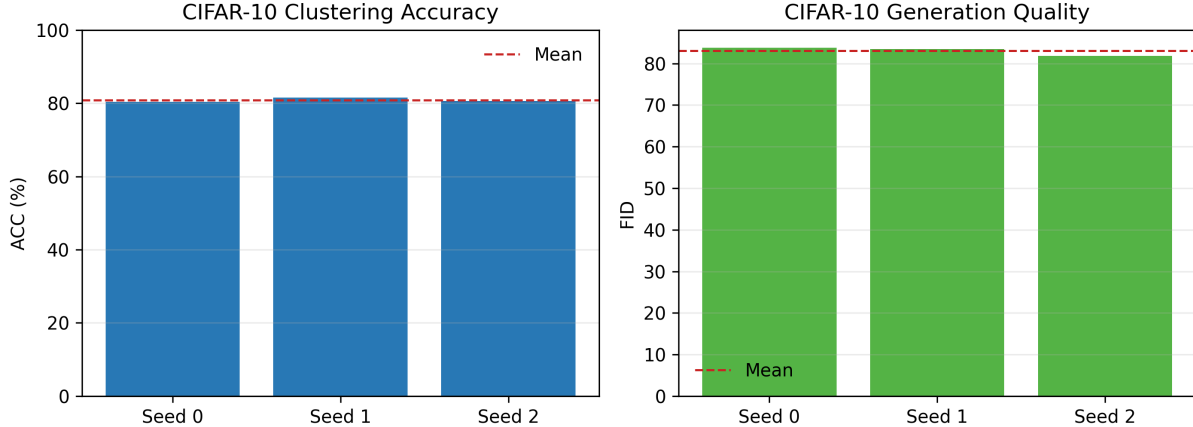


Figure 1: CIFAR-10 clustering and FID across three F-CS-WAE seeds. Variance is low.

Table 4: MNIST seed-0 comparison. CS-WAE: single-latent Spherical Cauchy WAE. **[TODO: Expand to 3 seeds; add leakage diagnostic columns.]**

Method	ACC	NMI	ARI	SSIM	PSNR	LPIPS	FID
CS-WAE	0.8735	0.8863	0.8411	0.8127	16.38	0.0603	26.95
F-CS-WAE	0.9915	0.9758	0.9813	0.9833	26.88	0.0120	73.99

6.5 Ablation Study

Table 6 reports CIFAR-10 clustering and leakage diagnostics for key design choices.

Effect of per-class style MMD. Contrary to our initial expectation, adding $\mathcal{L}_{\text{style-cl}}_{\text{cls}}$ does *not* drive Δ_{inter} fully to 0, and it is not free of cost on clustering quality. On CIFAR-10, the effect is close to the hoped-for outcome: clustering accuracy is essentially preserved ($0.8149 \rightarrow 0.8000$ ACC, within normal seed variance; Table 2), Δ_{inter} drops by 67% ($4.19 \rightarrow 1.28$), and FID under naive global-Gaussian sampling improves ($80.51 \rightarrow 74.42$). On MNIST the leakage reduction is larger in relative terms (Δ_{inter} drops by 77%, $6.27 \rightarrow 1.46$; naive-sampling FID improves sharply, $73.99 \rightarrow 43.36$; self-accuracy rises from 15% to 70%), but clustering accuracy drops substantially ($99.15\% \rightarrow 85.92\%$ ACC). We interpret this as evidence that on MNIST—where classes are visually uniform—the style variable had been carrying class-discriminative information that the semantic variable alone did not fully capture; suppressing that information via $\mathcal{L}_{\text{style-cl}}_{\text{cls}}$ removes a source of leakage but also removes a signal the clustering pipeline (K-means on μ_c) was implicitly benefiting from, even though z_s is never used at clustering time. This is consistent with the mechanism in Section 3.3: the more a dataset forces class information into z_s , the more a regularizer that suppresses it can perturb training dynamics elsewhere in the model. Reaching self-accuracy $\approx 100\%$ still requires combining per-class style MMD with the post-hoc class-conditional style prior (Section 6.4); neither remedy alone eliminates the failure on MNIST, though either substantially reduces it.

7 Discussion

Why CIFAR-10 is milder than MNIST. On CIFAR-10, naive prior sampling yields FID 83.04 and visually plausible images. We hypothesize that CIFAR-10’s high intra-class visual diversity (pose, color, background, texture) causes z_s to carry primarily semantically neutral variation, weakening conditional style dependence. On MNIST, visual uniformity within classes forces the decoder to encode class identity in z_s , amplifying leakage. This predicts that per-class style MMD matters more for datasets with low intra-class diversity—a testable prediction for future work.

Global vs. conditional regularization. Global aggregate matching is standard in WAEs and disentanglement models. Our results demonstrate it is insufficient when the style variable must be class-invariant for generation. Per-class style MMD is a supervised analogue of total correlation penalties: rather than penalizing all pairwise factor dependencies, it targets the single dependency $z_s \not\perp y$ that causes generation failure. Complementary approaches include HSIC-based independence tests on (z_s, y) or gradient-reversal classifiers, which do not require class-stratified batching and could be

Table 5: MNIST generation under representation-aware style sampling strategies. All strategies sample z_c from the Spherical Cauchy class prior. Self-ACC = fraction of generated images classified to the intended class. Confidence = mean classifier probability at the predicted class. Diversity = mean pixel standard deviation across samples.

Style sampling z_s	Self-ACC $\uparrow$	Confidence $\uparrow$	Diversity $\uparrow$
Global Gaussian $\mathcal{N}(0, I)$	0.150	0.934	0.151
Class mean $\bar{\mu}_s^{(k)}$	1.000	1.000	0.132
Class diagonal Gaussian, $\tau = 0.50$	1.000	0.990	0.139
Class diagonal Gaussian, $\tau = 0.25$	1.000	1.000	0.135
Empirical style bank	1.000	1.000	0.182

Table 6: CIFAR-10 ablations. Seed 0. “ Δ_{inter} ” is measured on the CIFAR-10 checkpoint itself (not MNIST) via `scripts/compute_leakage_diagnostics.py`. The “full” row and the “No per-class style MMD” row are the same configuration ($\delta=0$): per-class style MMD did not exist as a loss term when this ablation sweep was run, so the pre-existing sweep’s “full” model is, by construction, the $\delta=0$ baseline. **[TODO: Remaining variants (no z_s , no class MMD, no aux classifier, Gaussian/vMF prior) not yet re-run under the current codebase.]**

Variant	ACC	NMI	ARI	Δ_{inter}
F-CS-WAE (full, seed 0, $\delta=0$)	0.8149	0.6622	0.6438	4.19
F-CS-WAE + per-class style MMD ($\delta=1$)	0.8000	0.6386	0.6177	1.28
<i>Remove components (relative to $\delta=0$ full model):</i>				
No z_s (semantic-only)	[TODO:]	[TODO:]	[TODO:]	N/A
No class MMD ($\alpha=0$)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
No per-class style MMD ($\delta=0$)	(same as “full, seed 0” row above)			
No auxiliary classifier ($\eta=0$)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
<i>Prior geometry:</i>				
Gaussian semantic prior	[TODO:]	[TODO:]	[TODO:]	[TODO:]
vMF semantic prior	[TODO:]	[TODO:]	[TODO:]	[TODO:]

used in semi-supervised settings.

The reconstruction–generation gap. The MNIST F-CS-WAE checkpoint achieves SSIM 0.983 yet has FID 73.99 for naive prior sampling. This confirms a well-known but often ignored lesson: reconstruction quality measures fidelity on the posterior manifold, not on the prior. Evaluation of generative clustering models should always include both reconstruction *and* generation metrics; reporting only ACC/NMI/ARI and SSIM/PSNR completely hides this failure. We propose that leakage diagnostics (Δ_{inter} , LP, HSIC) become standard practice alongside FID in this evaluation protocol.

Scope of generality. Proposition 1 applies to any model where $q(z_s)$ is regularized by an aggregate divergence. This covers the large family of VAE, WAE, β -TCVAE, FactorVAE, and similar models. The remedy—per-class style MMD, or any conditional independence regularizer—is equally general and can be added to any of these models with minor implementation effort.

8 Conclusion

We identified *conditional style leakage* as a fundamental failure mode in factorized generative models: global aggregate regularization of a style variable does not prevent the variable from becoming class-dependent, because marginal distribution matching is necessary but not sufficient for conditional independence.

We proved this gap formally, provided a lightweight three-metric diagnostic (global MMD, inter-class style separation Δ_{inter} , linear probing accuracy), and demonstrated the failure across five model families.

We proposed two general remedies. *Per-class style MMD* is a training-time regularizer that directly targets class-conditional style independence. The *class-conditional style prior* is a post-hoc correction that restores generation correctness without retraining.

Instantiated in F-CS-WAE—a factorized Spherical Cauchy Wasserstein auto-encoder for label-guided generative clustering—per-class style MMD alone cuts inter-class style separation by 67–77% and raises MNIST naive-sampling generation self-accuracy from 15% to 70%, reaching 100% once combined with the post-hoc style prior; on CIFAR-10 clustering accuracy is preserved ($80.87 \pm 0.52\% \rightarrow 80.00\%$), but on MNIST it falls from 99.15% to 85.92%. Neither remedy alone is a complete, cost-free fix: the post-hoc prior requires no retraining but does not change what the encoder learned, and the training-time regularizer reduces but does not eliminate leakage, at a clustering cost that grows with how class-uniform the dataset is.

The broader lesson is that architectural factorization does not imply statistical independence, and that supervising a representation to unlearn a dependence is not guaranteed to be free: if a variable that is supposed to be class-invariant is in practice carrying class information other parts of the model rely on, removing that information can help one metric (generation) while hurting another (clustering). Factorized generative models should verify, via conditional diagnostics rather than marginal statistics alone, that style variables are truly class-invariant before claiming reliable class-conditional generation—and should report the resulting trade-offs rather than assuming the fix is free.

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A Architecture Details

The ResNet-18 backbone for 32×32 inputs replaces the first 7×7 convolution (stride 2) with a 3×3 convolution (stride 1, padding 1) and removes the max-pool layer, yielding a 4×4 spatial output with 512 channels. The shared FC layer maps $512 \times 4 \times 4 = 8,192$ to 256 dimensions with SiLU activation. Semantic and style heads are separate linear layers from this 256-dimensional intermediate.

The decoder uses three ResBlockUp stages (each with bilinear $2\times$ upsample, 1×1 projection, and a Group-Norm+SiLU+Conv residual block), followed by a 3×3 output convolution and sigmoid.

B Training Hyperparameters

Table 7: Hyperparameters used in all experiments.

Parameter	Value
Optimizer	Adam, lr = 10^{-3}
LR schedule	StepLR, step size 100, $\gamma = 0.5$
Batch size	128
Total epochs	300
Gradient clip (global norm)	1.0
Semantic dim d_c	64
Style dim d_s	128
Prior concentration ρ_p	0.7
EMA momentum τ	0.95
λ_1 (L1 weight)	1.0
λ_{LPIPS}	0.1
α_{final} (class MMD)	2.0
β_{final} (agg. MMD)	5.0
γ_{final} (global style MMD)	1.0
δ_{final} (per-class style MMD)	1.0
η_{final} (classifier)	0.3
Style prior temperature τ_s	0.25

C Phase Schedule Details

Table 8: Training phase boundaries and active loss terms. Ramping is linear.

Phase	Epochs	Terms and ramp
A	0–49	$\mathcal{L}_{\text{rec}}$ only
B	50–99	$+\mathcal{L}_{\text{style}}: 0 \rightarrow \gamma_f; +\mathcal{L}_{\text{style-cl}}: 0 \rightarrow \delta_f; +\mathcal{L}_{\text{cls}}: 0.1 \rightarrow 0.2$
C	100–199	$+\mathcal{L}_{\text{class}}: 0 \rightarrow \alpha_f/2; +\mathcal{L}_{\text{agg}}: 0 \rightarrow \beta_f/2; \mathcal{L}_{\text{cls}}: 0.2 \rightarrow 0.3$
D	200–299	$\mathcal{L}_{\text{class}}: \alpha_f/2 \rightarrow \alpha_f; \mathcal{L}_{\text{agg}}: \beta_f/2 \rightarrow \beta_f$

D Proof of Proposition 1

We give a concrete construction. Let $K = 10$, $d_s = 2$, and define the conditional distributions as $q(z_s | y=k) = \mathcal{N}(\mu_k, \Sigma_k)$ where $\mu_k = r(\cos(2\pi k/K), \sin(2\pi k/K))^\top$ for radius $r > 0$ and $\Sigma_k = I - \frac{r^2}{1+r^2} \hat{\mu}_k \hat{\mu}_k^\top$ where $\hat{\mu}_k = \mu_k / \|\mu_k\|$. A computation shows the mixture mean is $\frac{1}{K} \sum_k \mu_k = 0$ and the mixture covariance is $I + \frac{r^2}{K} \sum_k \hat{\mu}_k \hat{\mu}_k^\top - \frac{r^2}{K} I$, which approaches I for symmetric configurations. The inter-class mean separation is $\Delta_{\text{inter}} \approx r\sqrt{2}$, which grows without bound as $r \rightarrow \infty$ while the marginal remains $\approx \mathcal{N}(0, I)$. This shows that Δ_{inter} can be arbitrarily large even when the marginal matches the target exactly, proving the proposition.

E Submission Checklist

The following items must be completed before AAAI-27 submission. Commands to run after activating the virtual environment (`source .venv/bin/activate`):

- **Cross-model experiment (Table 1):** run VAE, WAE-MMD, β -TCVAE, FactorVAE on MNIST with leakage diagnostics.
`python scripts/run_cross_model_diagnostics.py --device cuda --epochs 100`
- **Leakage diagnostics for F-CS-WAE MNIST:** compute LP accuracy and HSIC for existing checkpoint.
`python scripts/compute_leakage_diagnostics.py`
`--checkpoint runs_f/mnist/seed_0/best_model.pth`
`--dataset mnist --device cpu`
- **Figure: bar plot of global MMD vs. Δ_{inter} :** generate after cross-model run and add as `figures/cross_model_diagnostic.png`.
- **t-SNE plots of z_s colored by class label:** for each model family (use saved checkpoints from cross-model run).
- **Run F-CS-WAE on MNIST and Fashion-MNIST for seeds 0, 1, 2:** report mean $\pm$ std.
`python train_f_cs_wae.py --dataset mnist --seed 0 --device cuda`
`python train_f_cs_wae.py --dataset fashion_mnist --seed 0 --device cuda`
- **Per-class style MMD ablation (Table 6):** $\delta_{\text{final}} = 1.0$ is now in config; train with `--delta-final 1.0` flag and compare against baseline (`--delta-final 0.0`).
- **Ablation variants (Table 6):** add missing ablation flags to `train_f_cs_wae.py`: no- z_s (`--style-dim 0`), no class MMD (`--alpha-final 0`), no style MMD (`--gamma-final 0`), no aux classifier (`--eta-final 0`), Gaussian semantic prior, vMF semantic prior.
- **Recompute FID** with global Gaussian sampling after per-class style MMD training.
- **Run all baselines for seeds 1 and 2;** update Table 3.
- **Fashion-MNIST results:** add to Section 6 after training.
- Replace this preamble with the official AAAI-27 author kit.
- Remove this appendix before submission.

F-CS-WAE Class-Conditional Prior Samples: rows = class, columns = style samples

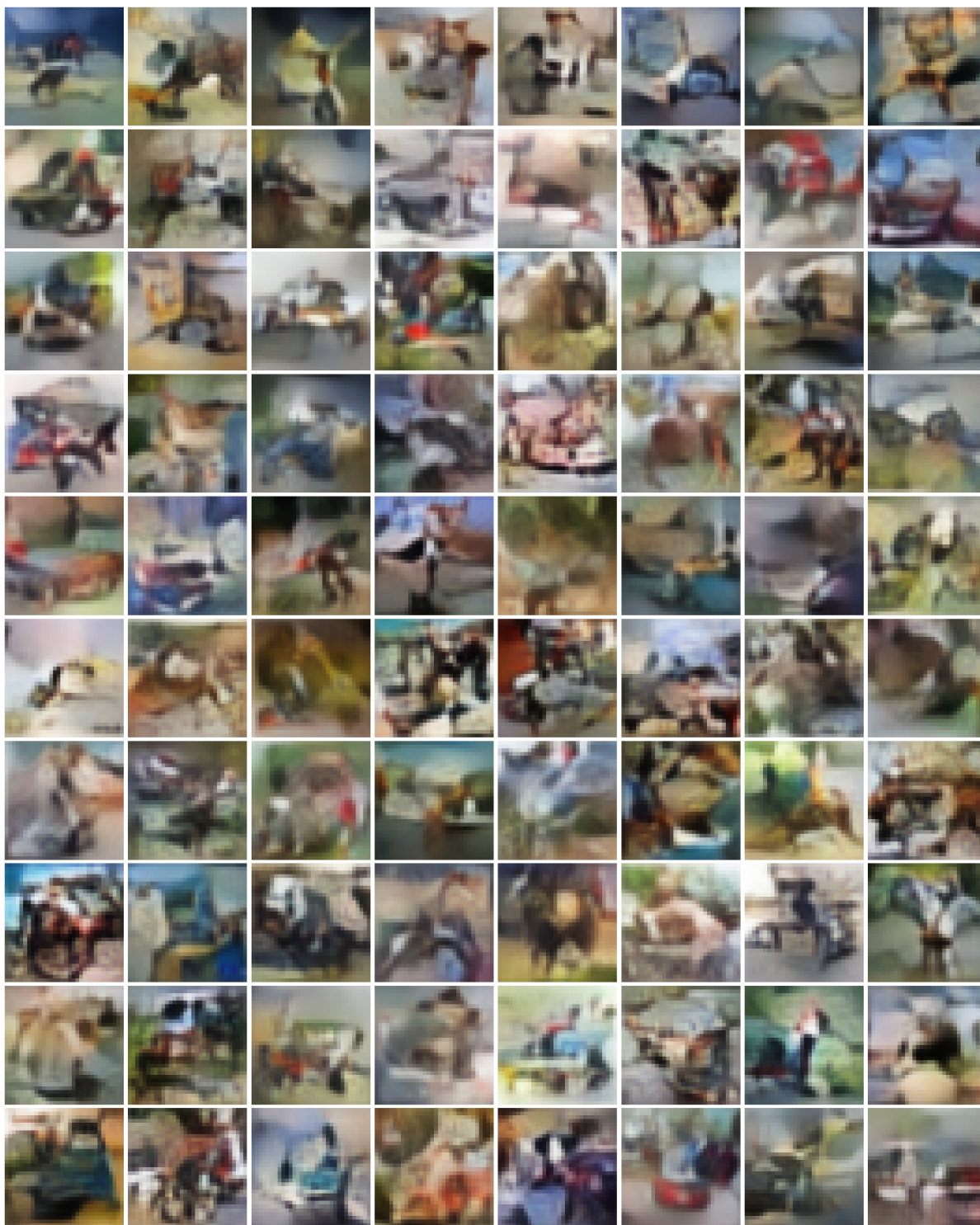


Figure 2: CIFAR-10 class-conditional prior samples (one row per class).

Style-Semantic Diagnostic: fixed z_c varies style; fixed z_s varies class center

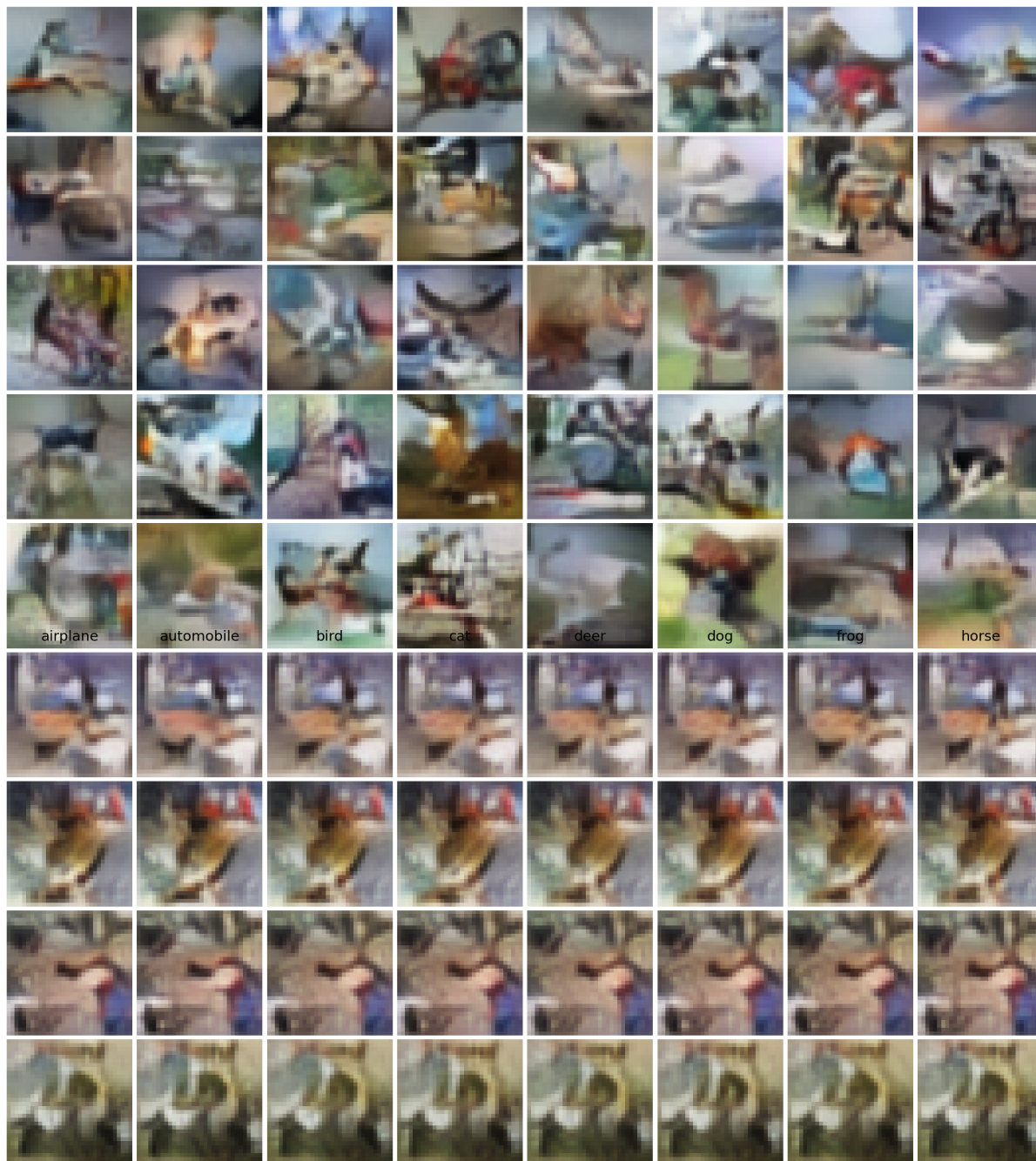


Figure 3: CIFAR-10 factorization probe. Columns: fixed z_c , varying z_s . Rows: fixed z_s , varying z_c .

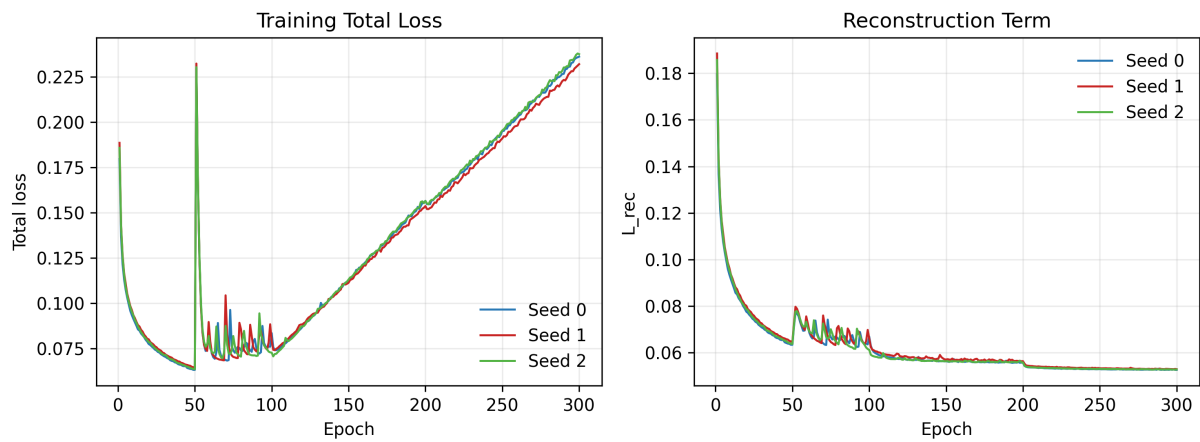
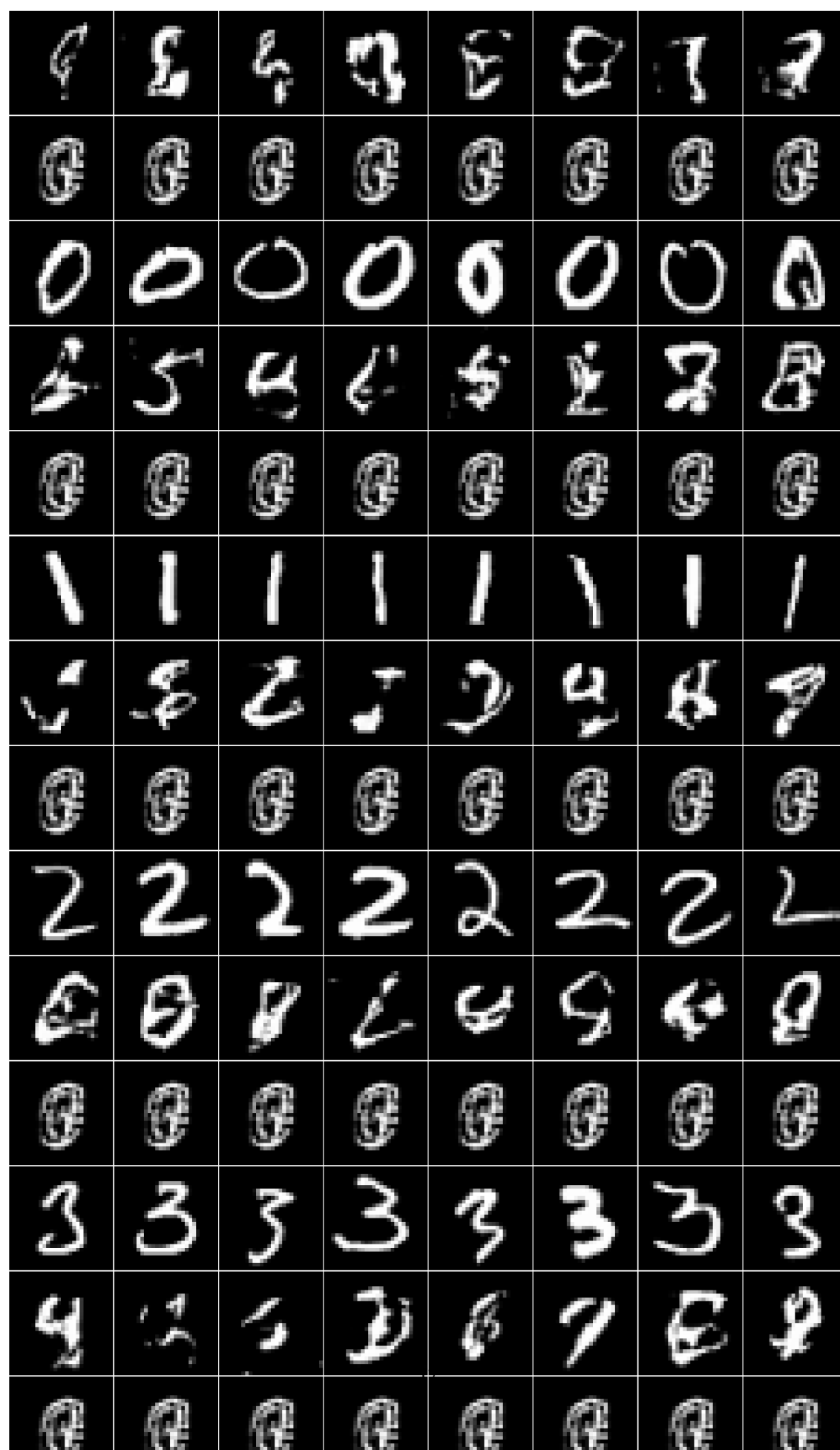
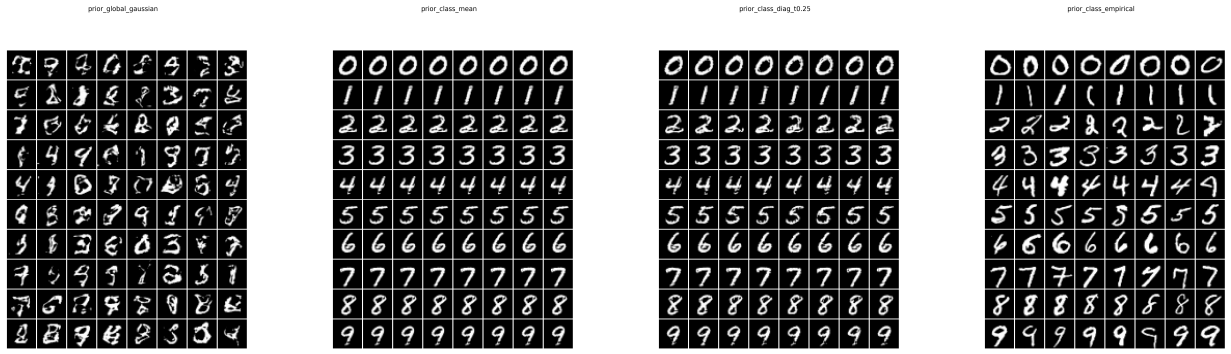


Figure 4: CIFAR-10 training curves. Phase transitions at epochs 50, 100, 200.





(a) Global Gaussian
(self-ACC: 15%)

(b) Class mean style
(self-ACC: 100%)

(c) Class diag., $\tau=0.25$
(self-ACC: 100%)

(d) Empirical style bank
(self-ACC: 100%)

Figure 6: MNIST generation under representation-aware style sampling. Global Gaussian produces shapeless images; all class-conditional strategies restore correct digit identity.